Game Design Documentation

October 21, 2026

Part 1: Engine

1 Engine entry point: The life cycle of the game.

1.1 Entry Point. File: EntryPoint.c

Step1 Program execution: Player click the game and execute it.

Step2 Allocate memories for every units: Sprites, Renderer.

Step3 Register Windows.

Step4 Initialize rendering program.

Step5 **Begin the windows loop.** (The core of gaming loops. More details, see 1.2).

Step6 rendering shutdown.

Step7 free_memories.

1.2 Frame Loop. File: HWND.c

Step1 Program state checking: If program running state is true, then begin the frame loop.

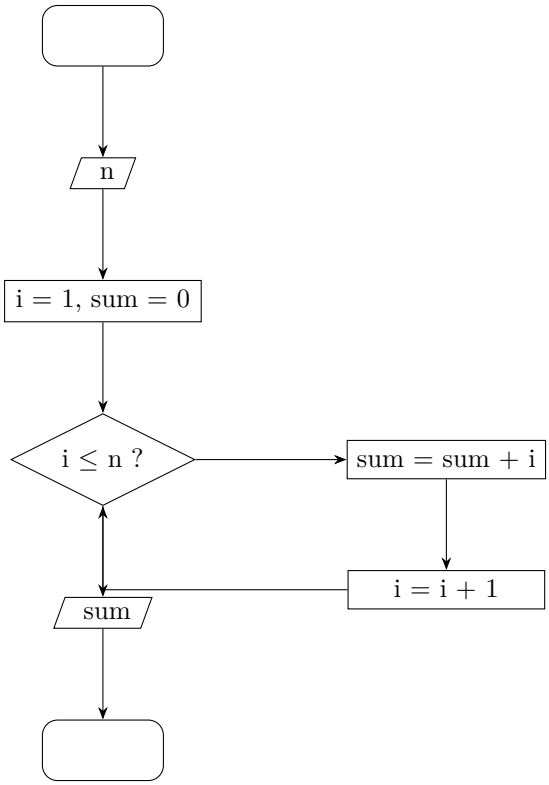
Step2 Peek Message (Windows API): Detect whether player pressed the Key code, this API will help us to detect the key code instantly.

Step3 **Update rendering:** when peek message finished, then execute the rendering update. Draw the current prepared frame. More details, see **??**.

Step4 Time counter: We use frame to define the game's minimum time. The time counter defined in the Timer.c file.

Step5 Back to the Step 1. Check whether the program running, if running, then prepare for the next Peek Message and new frame drawing. If not, then shut down the frame.

1.3 Update: Flow Chart.



2 Engine Initialization

1. Allocate memories for sprites and musics.
2. Initialize the sprite path so that find quickly.
3. Allocate memories for display.
4. Call Windows HWND programme. See 4.1.

3 Render

1.

4 Details

4.1 System Adaptive

Windows:

1. Register the windows.
2. Create the windows.
3. Loop the windows.
4. Destory the windows.

4.2 Memory Manager

The engine system allocate memories for sprites and musics, then the release of the sprites and musics are managed by themselves.